**The University of the State of New York**
**REGENTS HIGH SCHOOL EXAMINATION**

# GEOMETRY

**Tuesday,** June 19, 2018 — 9:15 a.m. to 12:15 p.m., only

**Student Name:** ______________________________________________

**School Name:** ______________________________________________

**The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.**

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 35 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III,** and **IV** directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

**Notice…**
**A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.**

**DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.**

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## High School Math Reference Sheet

| | | |
|---|---|---|
| 1 inch = 2.54 centimeters | 1 kilometer = 0.62 mile | 1 cup = 8 fluid ounces |
| 1 meter = 39.37 inches | 1 pound = 16 ounces | 1 pint = 2 cups |
| 1 mile = 5280 feet | 1 pound = 0.454 kilogram | 1 quart = 2 pints |
| 1 mile = 1760 yards | 1 kilogram = 2.2 pounds | 1 gallon = 4 quarts |
| 1 mile = 1.609 kilometers | 1 ton = 2000 pounds | 1 gallon = 3.785 liters |
| | | 1 liter = 0.264 gallon |
| | | 1 liter = 1000 cubic centimeters |

| | |
|---|---|
| Triangle | $A = \frac{1}{2}bh$ |
| Parallelogram | $A = bh$ |
| Circle | $A = \pi r^2$ |
| Circle | $C = \pi d$ or $C = 2\pi r$ |
| General Prisms | $V = Bh$ |
| Cylinder | $V = \pi r^2 h$ |
| Sphere | $V = \frac{4}{3}\pi r^3$ |
| Cone | $V = \frac{1}{3}\pi r^2 h$ |
| Pyramid | $V = \frac{1}{3}Bh$ |

| | |
|---|---|
| Pythagorean Theorem | $a^2 + b^2 = c^2$ |
| Quadratic Formula | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ |
| Arithmetic Sequence | $a_n = a_1 + (n - 1)d$ |
| Geometric Sequence | $a_n = a_1 r^{n-1}$ |
| Geometric Series | $S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$ |
| Radians | 1 radian $= \frac{180}{\pi}$ degrees |
| Degrees | 1 degree $= \frac{\pi}{180}$ radians |
| Exponential Growth/Decay | $A = A_0 e^{k(t - t_0)} + B_0$ |

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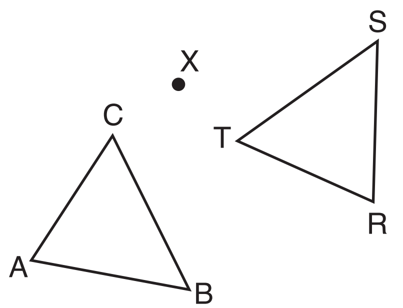
## Part I

**Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet.** [48]

**Use this space for computations.**

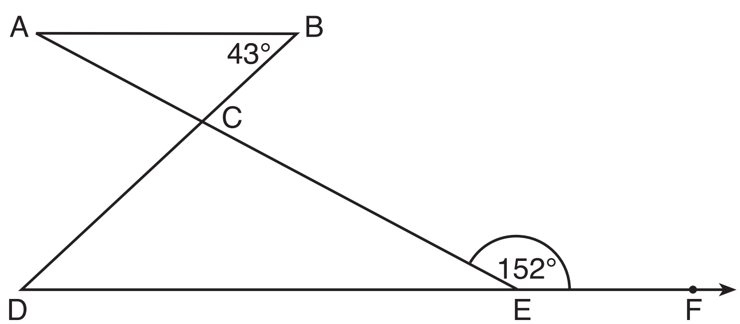
**1** After a counterclockwise rotation about point $X$, scalene triangle $ABC$ maps onto $\triangle RST$, as shown in the diagram below.




Which statement must be true?

(1) $\angle A \cong \angle R$
(2) $\angle A \cong \angle S$
(3) $\overline{CB} \cong \overline{TR}$
(4) $\overline{CA} \cong \overline{TS}$

**2** In the diagram below, $\overline{AB} \parallel \overrightarrow{DEF}$, $\overline{AE}$ and $\overline{BD}$ intersect at $C$, $m\angle B = 43°$, and $m\angle CEF = 152°$.

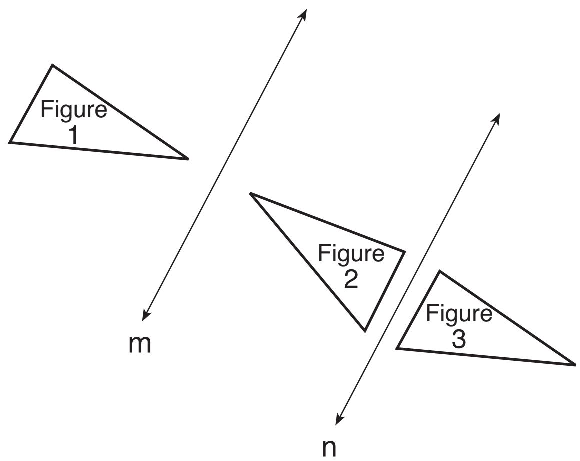



Which statement is true?

(1) $m\angle D = 28°$
(2) $m\angle A = 43°$
(3) $m\angle ACD = 71°$
(4) $m\angle BCE = 109°$

**Use this space for computations.**

**3** In the diagram below, line $m$ is parallel to line $n$. Figure 2 is the image of Figure 1 after a reflection over line $m$. Figure 3 is the image of Figure 2 after a reflection over line $n$.




Which single transformation would carry Figure 1 onto Figure 3?

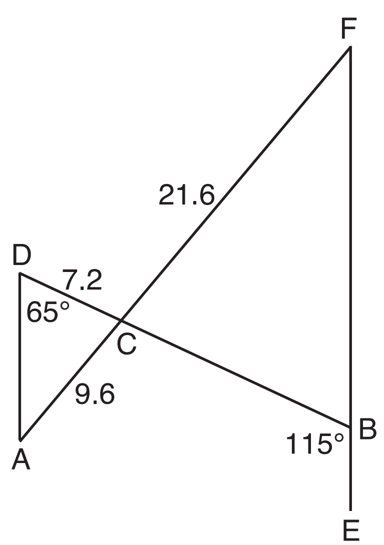
(1) a dilation

(2) a rotation

(3) a reflection

(4) a translation

**4** In the diagram below, $\overline{AF}$ and $\overline{DB}$ intersect at C, and $\overline{AD}$ and $\overline{FBE}$ are drawn such that $m\angle D = 65°$, $m\angle CBE = 115°$, $DC = 7.2$, $AC = 9.6$, and $FC = 21.6$.




What is the length of $\overline{CB}$?

(1) 3.2

(2) 4.8

(3) 16.2

(4) 19.2

**Use this space for computations.**

**5** Given square $RSTV$, where $RS = 9$ cm. If square $RSTV$ is dilated by a scale factor of 3 about a given center, what is the perimeter, in centimeters, of the image of $RSTV$ after the dilation?

(1) 12 (3) 36

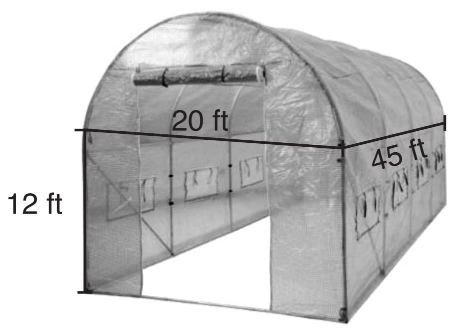
(2) 27 (4) 108

**6** In right triangle $ABC$, hypotenuse $\overline{AB}$ has a length of 26 cm, and side $\overline{BC}$ has a length of 17.6 cm. What is the measure of angle $B$, to the *nearest degree*?

(1) 48° (3) 43°

(2) 47° (4) 34°

**7** The greenhouse pictured below can be modeled as a rectangular prism with a half-cylinder on top. The rectangular prism is 20 feet wide, 12 feet high, and 45 feet long. The half-cylinder has a diameter of 20 feet.




To the *nearest cubic foot*, what is the volume of the greenhouse?

(1) 17,869 (3) 39,074

(2) 24,937 (4) 67,349

**Use this space for computations.**

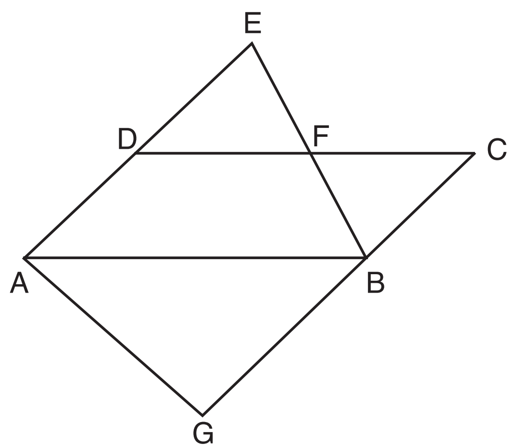
**8** In a right triangle, the acute angles have the relationship $\sin(2x + 4) = \cos(46)$.

What is the value of $x$?

(1) 20 (3) 24

(2) 21 (4) 25

**9** In the diagram below, $\overline{AB} \parallel \overline{DFC}$, $\overline{EDA} \parallel \overline{CBG}$, and $\overline{EFB}$ and $\overline{AG}$ are drawn.




Which statement is always true?

(1) $\triangle DEF \cong \triangle CBF$ (3) $\triangle BAG \sim \triangle AEB$

(2) $\triangle BAG \cong \triangle BAE$ (4) $\triangle DEF \sim \triangle AEB$

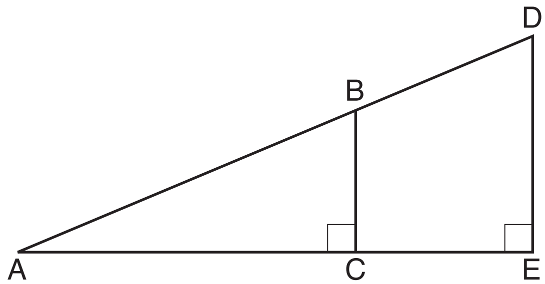
**10** The base of a pyramid is a rectangle with a width of 4.6 cm and a length of 9 cm. What is the height, in centimeters, of the pyramid if its volume is 82.8 cm$^3$?

(1) 6 (3) 9

(2) 2 (4) 18

**Use this space for computations.**

**11** In the diagram below of right triangle $AED$, $\overline{BC} \parallel \overline{DE}$.




Which statement is always true?

(1) $\frac{AC}{BC} = \frac{DE}{AE}$

(2) $\frac{AB}{AD} = \frac{BC}{DE}$

(3) $\frac{AC}{CE} = \frac{BC}{DE}$

(4) $\frac{DE}{BC} = \frac{DB}{AB}$

**12** What is an equation of the line that passes through the point (6,8) and is perpendicular to a line with equation $y = \frac{3}{2}x + 5$?

(1) $y - 8 = \frac{3}{2}(x - 6)$

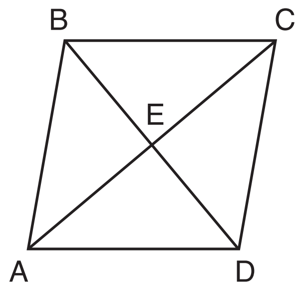
(2) $y - 8 = -\frac{2}{3}(x - 6)$

(3) $y + 8 = \frac{3}{2}(x + 6)$

(4) $y + 8 = -\frac{2}{3}(x + 6)$

**Use this space for computations.**

**13** The diagram below shows parallelogram $ABCD$ with diagonals $\overline{AC}$ and $\overline{BD}$ intersecting at $E$.




What additional information is sufficient to prove that parallelogram $ABCD$ is also a rhombus?

(1) $\overline{BD}$ bisects $\overline{AC}$.

(2) $\overline{AB}$ is parallel to $\overline{CD}$.

(3) $\overline{AC}$ is congruent to $\overline{BD}$.

(4) $\overline{AC}$ is perpendicular to $\overline{BD}$.

**14** Directed line segment $DE$ has endpoints $D(-4,-2)$ and $E(1,8)$. Point $F$ divides $\overline{DE}$ such that $DF:FE$ is 2:3. What are the coordinates of $F$?

(1) $(-3,0)$

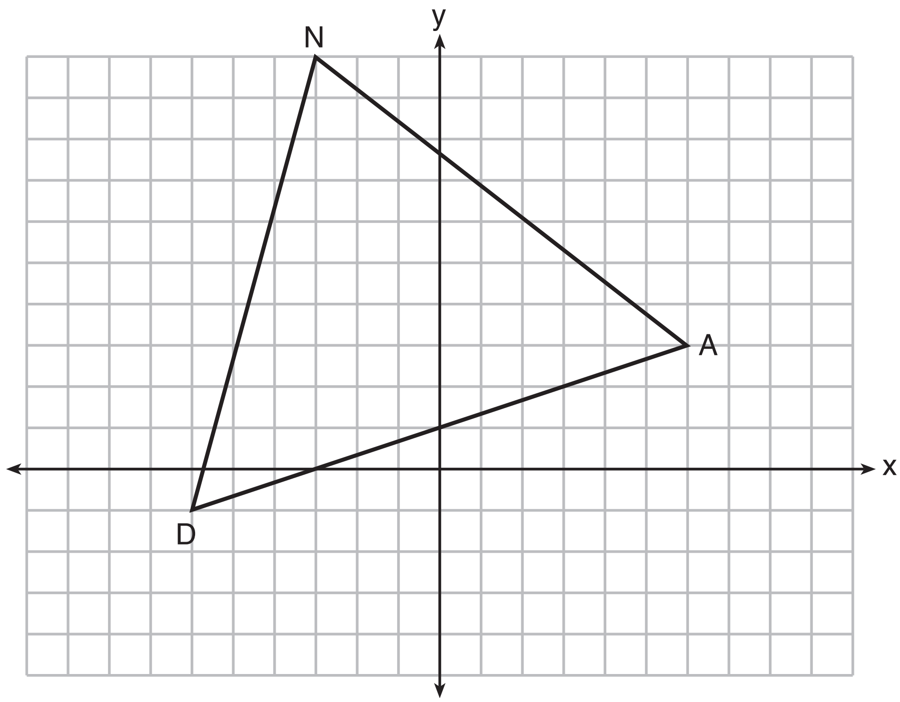
(2) $(-2,2)$

(3) $(-1,4)$

(4) $(2,4)$

**Use this space for computations.**

**15** Triangle $DAN$ is graphed on the set of axes below. The vertices of $\triangle DAN$ have coordinates $D(-6,-1)$, $A(6,3)$, and $N(-3,10)$.




What is the area of $\triangle DAN$?

(1) 60

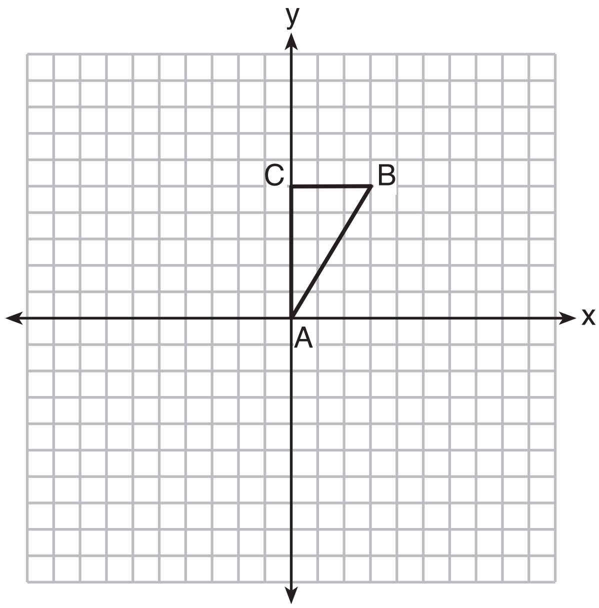
(2) 120

(3) $20\sqrt{13}$

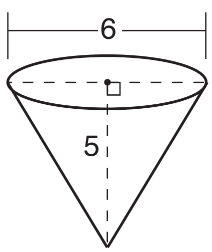
(4) $40\sqrt{13}$

**Use this space for computations.**

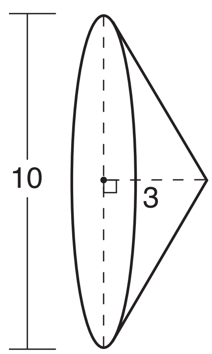
**16** Triangle $ABC$, with vertices at $A(0,0)$, $B(3,5)$, and $C(0,5)$, is graphed on the set of axes shown below.



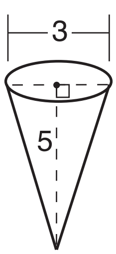

Which figure is formed when $\triangle ABC$ is rotated continuously about $\overline{BC}$?



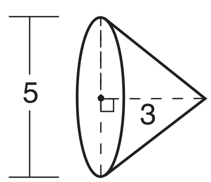

(1)




(3)



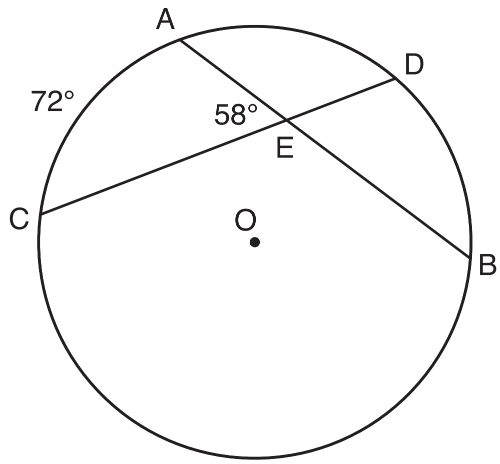

(2)




(4)

**Use this space for computations.**

**17** In the diagram below of circle $O$, chords $\overline{AB}$ and $\overline{CD}$ intersect at $E$.

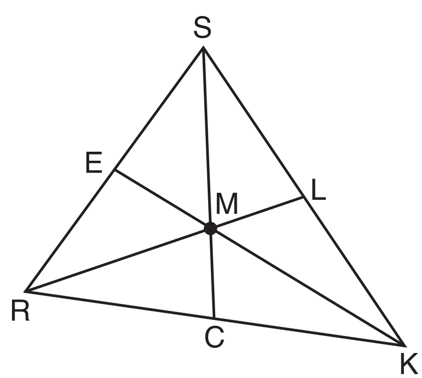



If $m\widehat{AC} = 72°$ and $m\angle AEC = 58°$, how many degrees are in $m\widehat{DB}$?

(1) 108° (3) 44°

(2) 65° (4) 14°

**18** In triangle $SRK$ below, medians $\overline{SC}$, $\overline{KE}$, and $\overline{RL}$ intersect at $M$.



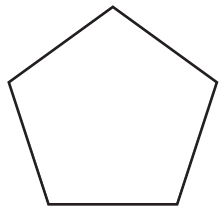

Which statement must always be true?

(1) $3(MC) = SC$ (3) $RM = 2MC$

(2) $MC = \frac{1}{3}(SM)$ (4) $SM = KM$

Use this space for computations.

**19** The regular polygon below is rotated about its center.

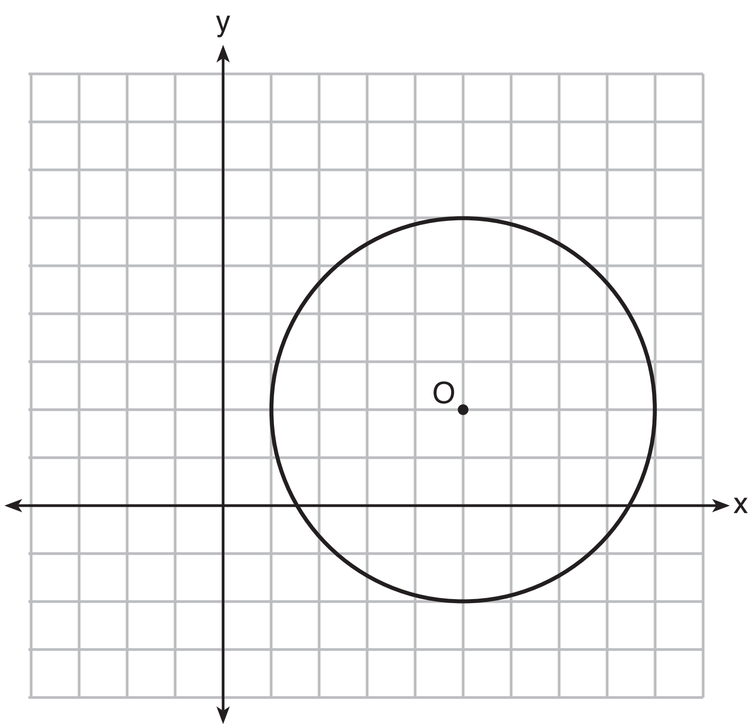


Which angle of rotation will carry the figure onto itself?

(1) 60° (3) 216°

(2) 108° (4) 540°

**20** What is an equation of circle $O$ shown in the graph below?

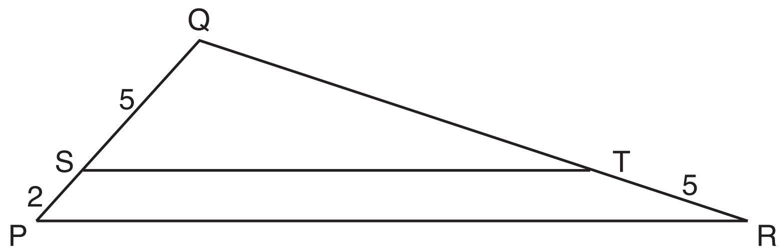



(1) $x^2 + 10x + y^2 + 4y = -13$ (3) $x^2 + 10x + y^2 + 4y = -25$

(2) $x^2 - 10x + y^2 - 4y = -13$ (4) $x^2 - 10x + y^2 - 4y = -25$

**Use this space for computations.**

**21** In the diagram below of $\triangle PQR$, $\overline{ST}$ is drawn parallel to $\overline{PR}$, $PS = 2$, $SQ = 5$, and $TR = 5$.

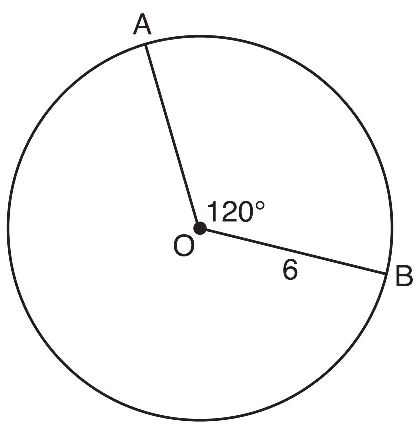



What is the length of $\overline{QR}$ ?

(1) 7 (3) $12\frac{1}{2}$

(2) 2 (4) $17\frac{1}{2}$

**22** The diagram below shows circle $O$ with radii $\overline{OA}$ and $\overline{OB}$. The measure of angle $AOB$ is 120°, and the length of a radius is 6 inches.




Which expression represents the length of arc $AB$, in inches?

(1) $\frac{120}{360}(6\pi)$ (3) $\frac{1}{3}(36\pi)$

(2) $120(6)$ (4) $\frac{1}{3}(12\pi)$

**Use this space for computations.**

**23** Line segment $CD$ is the altitude drawn to hypotenuse $\overline{EF}$ in right triangle $ECF$. If $EC = 10$ and $EF = 24$, then, to the *nearest tenth*, $ED$ is

(1) 4.2
(2) 5.4
(3) 15.5
(4) 21.8

**24** Line $MN$ is dilated by a scale factor of 2 centered at the point (0,6). If $\overleftrightarrow{MN}$ is represented by $y = -3x + 6$, which equation can represent $\overleftrightarrow{M'N'}$, the image of $\overleftrightarrow{MN}$?

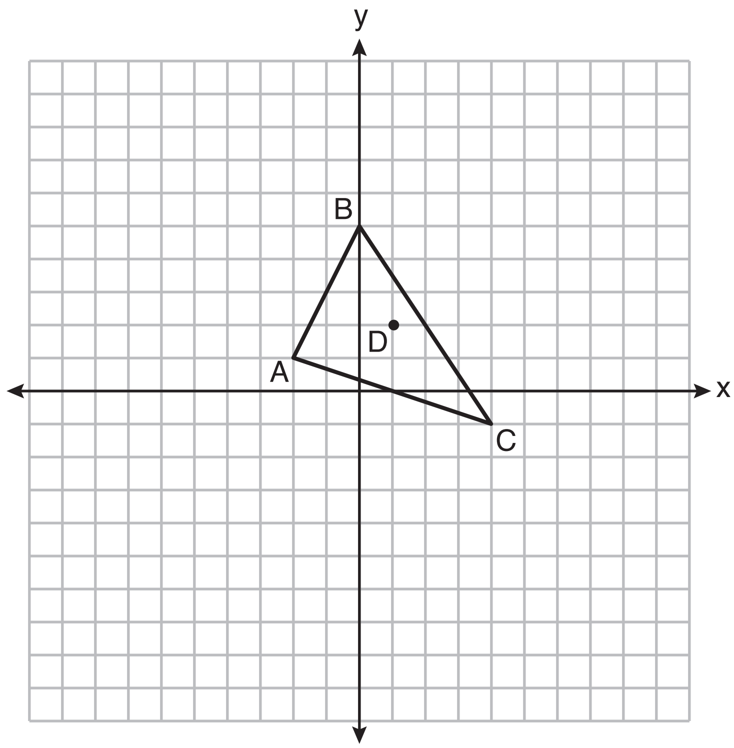
(1) $y = -3x + 12$
(2) $y = -3x + 6$
(3) $y = -6x + 12$
(4) $y = -6x + 6$

## Part II

**Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.** [14]

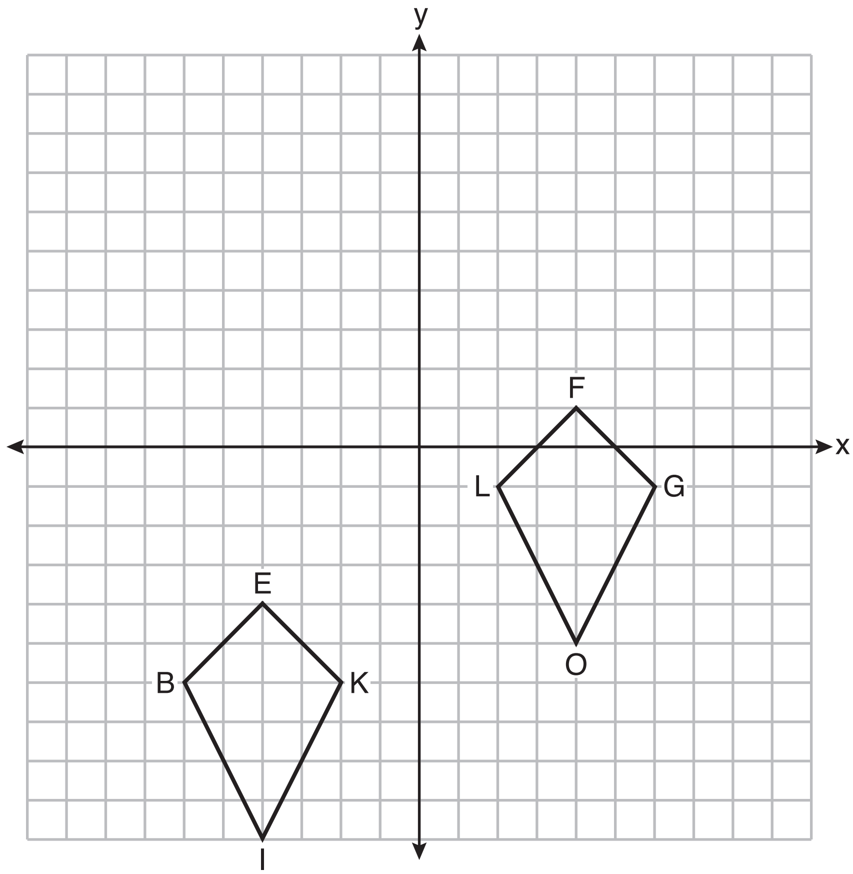
**25** Triangle $A'B'C'$ is the image of triangle $ABC$ after a translation of 2 units to the right and 3 units up. Is triangle $ABC$ congruent to triangle $A'B'C'$? Explain why.

**26** Triangle $ABC$ and point $D(1,2)$ are graphed on the set of axes below.



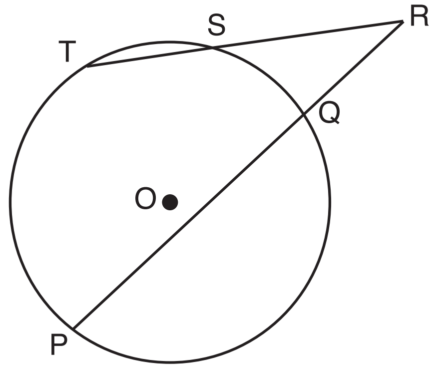

Graph and label $\triangle A'B'C'$, the image of $\triangle ABC$, after a dilation of scale factor 2 centered at point $D$.

**27** Quadrilaterals *BIKE* and *GOLF* are graphed on the set of axes below.



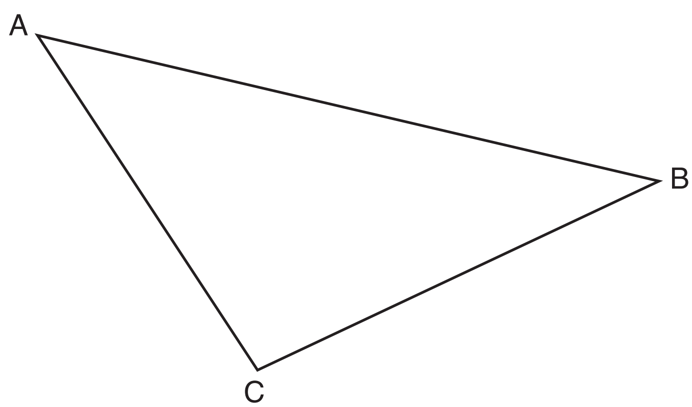

Describe a sequence of transformations that maps quadrilateral *BIKE* onto quadrilateral *GOLF*.

**28** In the diagram below, secants $\overline{RST}$ and $\overline{RQP}$, drawn from point $R$, intersect circle $O$ at $S$, $T$, $Q$, and $P$.

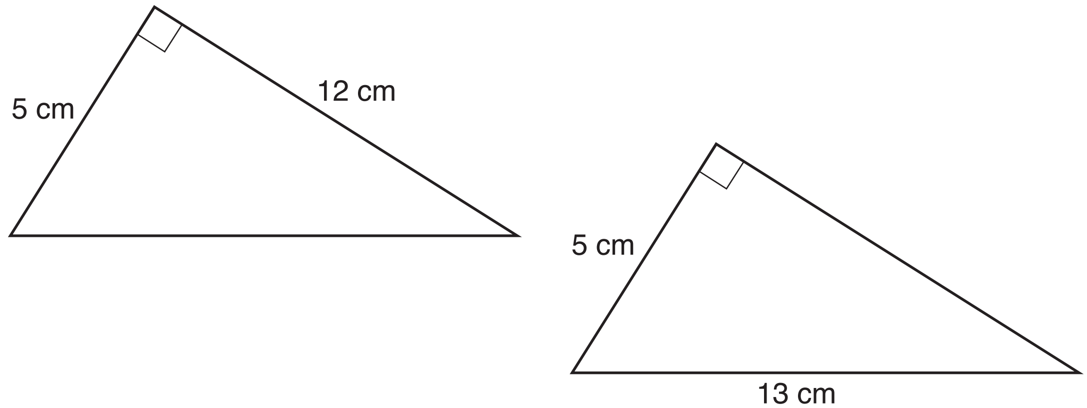



If $RS = 6$, $ST = 4$, and $RP = 15$, what is the length of $\overline{RQ}$?

**29** Using a compass and straightedge, construct the median to side $\overline{AC}$ in $\triangle ABC$ below. [Leave all construction marks.]



**30** Skye says that the two triangles below are congruent. Margaret says that the two triangles are similar.



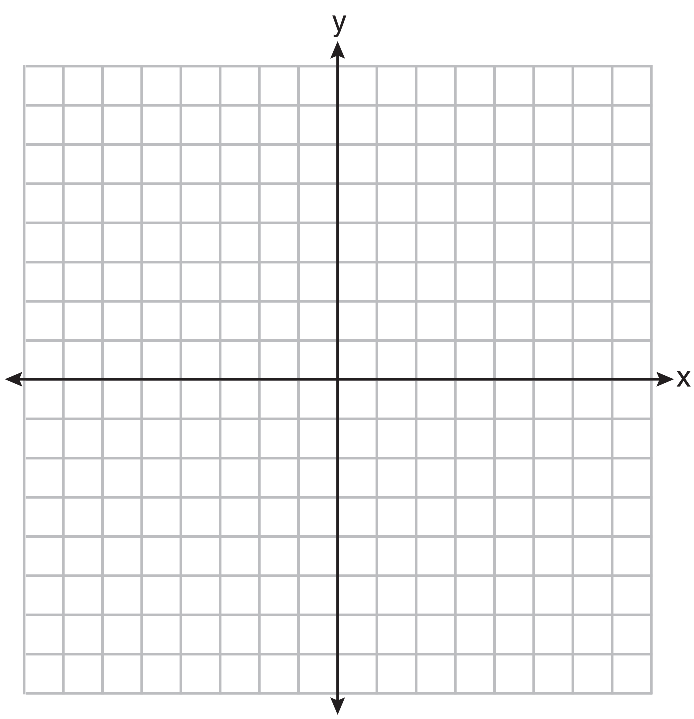

Are Skye and Margaret both correct? Explain why.

**31** Randy's basketball is in the shape of a sphere with a maximum circumference of 29.5 inches. Determine and state the volume of the basketball, to the *nearest cubic inch*.

## Part III

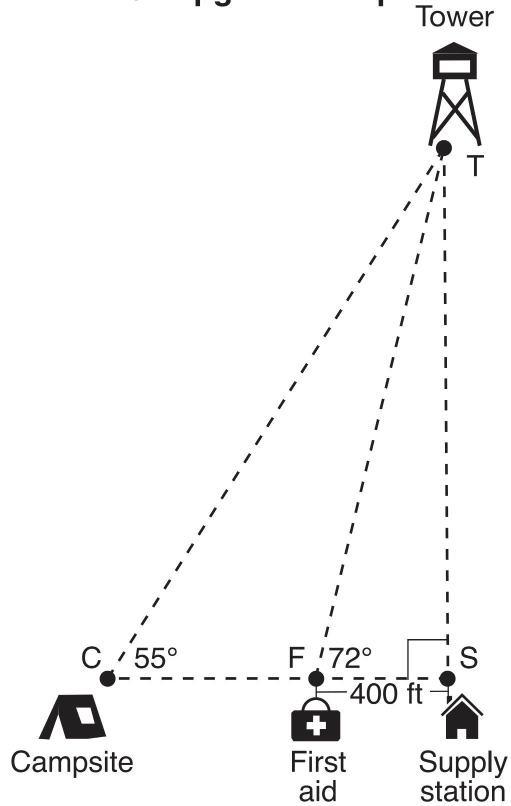
**Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.** [12]

**32** Triangle $ABC$ has vertices with coordinates $A(-1,-1)$, $B(4,0)$, and $C(0,4)$. Prove that $\triangle ABC$ is an isosceles triangle but *not* an equilateral triangle. [The use of the set of axes below is optional.]



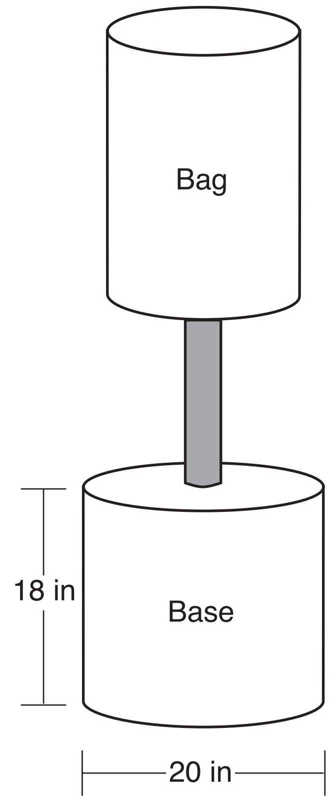
**33** The map of a campground is shown below. Campsite $C$, first aid station $F$, and supply station $S$ lie along a straight path. The path from the supply station to the tower, $T$, is perpendicular to the path from the supply station to the campsite. The length of path $\overline{FS}$ is 400 feet. The angle formed by path $\overline{TF}$ and path $\overline{FS}$ is 72°. The angle formed by path $\overline{TC}$ and path $\overline{CS}$ is 55°.

**Campground Map**




Determine and state, to the *nearest foot*, the distance from the campsite to the tower.

**34** Shae has recently begun kickboxing and purchased training equipment as modeled in the diagram below. The total weight of the bag, pole, and unfilled base is 270 pounds. The cylindrical base is 18 inches tall with a diameter of 20 inches. The dry sand used to fill the base weighs 95.46 lbs per cubic foot.

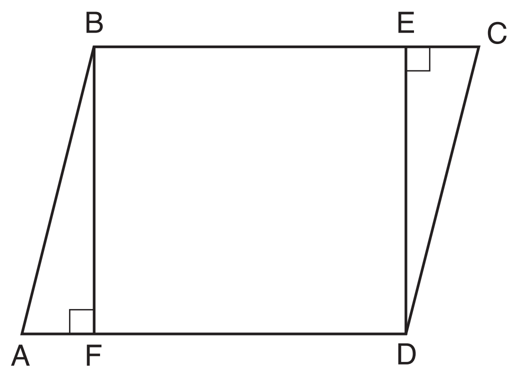



To the *nearest pound*, determine and state the total weight of the training equipment if the base is filled to 85% of its capacity.

## Part IV

**Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for the question to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.** [6]

**35** Given: Parallelogram $ABCD$, $\overline{BF} \perp \overline{AFD}$, and $\overline{DE} \perp \overline{BEC}$




Prove: $BEDF$ is a rectangle

**Work space for question 35 is continued on the next page.**

**If the student's responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.**

## Part I

Allow a total of 48 credits, 2 credits for each of the following. Allow credit if the student has written the correct answer instead of the numeral 1, 2, 3, or 4.

| | | |
|---|---|---|
| (1) . . . . . 1 . . . . . | (9) . . . . . 4 . . . . . | (17) . . . . . 3 . . . . . |
| (2) . . . . . 3 . . . . . | (10) . . . . . 1 . . . . . | (18) . . . . . 1 . . . . . |
| (3) . . . . . 4 . . . . . | (11) . . . . . 2 . . . . . | (19) . . . . . 3 . . . . . |
| (4) . . . . . 3 . . . . . | (12) . . . . . 2 . . . . . | (20) . . . . . 2 . . . . . |
| (5) . . . . . 4 . . . . . | (13) . . . . . 4 . . . . . | (21) . . . . . 4 . . . . . |
| (6) . . . . . 2 . . . . . | (14) . . . . . 2 . . . . . | (22) . . . . . 4 . . . . . |
| (7) . . . . . 1 . . . . . | (15) . . . . . 1 . . . . . | (23) . . . . . 1 . . . . . |
| (8) . . . . . 1 . . . . . | (16) . . . . . 3 . . . . . | (24) . . . . . 2 . . . . . |

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: http://www.p12.nysed.gov/assessment/ and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the "Model Response Set," for the Regents Examination in Geometry. This guidance is intended to be part of the scorer training. Schools should use the Model Response Set along with the rubrics in the Scoring Key and Rating Guide to help guide scoring of student work. While not reflective of all scenarios, the Model Response Set illustrates how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department's web site at: http://www.nysedregents.org/geometryre/.

## Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) **[2]** Yes, and a correct explanation is written.

**[1]** An appropriate explanation is written, but one conceptual error is made.

***or***

**[1]** An incomplete or partially correct explanation is written.

**[0]** Yes, but no explanation is written.

***or***

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) **[2]** Triangle $A'B'C'$ is graphed and labeled correctly.

**[1]** Appropriate work is shown, but one graphing error is made.

***or***

**[1]** One conceptual error is made, but appropriate vertices are graphed and labeled.

***or***

**[1]** The image of $\triangle ABC$ is graphed correctly, but it is not labeled or is labeled incorrectly.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(27) **[2]** A correct sequence of transformations is described.

**[1]** An appropriate sequence is described, but one graphing error is made.

***or***

**[1]** An appropriate sequence is described, but one conceptual error is made.

***or***

**[1]** A reflection and translation are identified, but no specific description is written.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(28) **[2]** 4, and correct work is shown.

**[1]** Appropriate work is shown, but one computational error is made.

***or***

**[1]** Appropriate work is shown, but one conceptual error is made.

***or***

**[1]** 4, but no work is shown.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(29) **[2]** A correct construction is drawn showing all appropriate arcs, and the median to $\overline{AC}$ is drawn.

**[1]** Appropriate work is drawn showing all appropriate arcs, but the median to $\overline{AB}$ or $\overline{BC}$ is drawn.

***or***

**[1]** A correct construction is drawn showing all appropriate arcs, but the median to $\overline{AC}$ is not drawn.

**[0]** A drawing that is not an appropriate construction is shown.

***or***

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(30) **[2]** Yes, and a correct explanation is written.

**[1]** An appropriate explanation is written, but one conceptual error is made.

***or***

**[1]** An incomplete or partially correct explanation is written.

**[0]** Yes, but the explanation is missing or incorrect.

***or***

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(31) **[2]** 433 or 434, and correct work is shown.

**[1]** Appropriate work is shown, but one computational or rounding error is made.

***or***

**[1]** Appropriate work is shown, but one conceptual error is made.

***or***

**[1]** 433 or 434, but no work is shown.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

## Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(32) **[4]** Correct work is shown to prove $\triangle ABC$ is an isosceles triangle and not an equilateral triangle, and correct concluding statements are made.

**[3]** Appropriate work is shown, but one computational or graphing error is made. Appropriate concluding statements are made.

***or***

**[3]** Appropriate work is shown to prove $\triangle ABC$ is an isosceles triangle, and a correct concluding statement is made, but no further correct work is shown.

**[2]** Appropriate work is shown, but two or more computational or graphing errors are made. Appropriate concluding statements are made.

***or***

**[2]** Appropriate work is shown, but one conceptual error is made. Appropriate concluding statements are made.

***or***

**[2]** Appropriate work is shown to find the lengths of all three sides, but no further correct work is shown.

**[1]** Appropriate work is shown to find the lengths of the two sides, but no further correct work is shown.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(33) **[4]** 1503, and correct work is shown.

**[3]** Appropriate work is shown, but one computational error is made.

**[2]** Appropriate work is shown, but two or more computational or rounding errors are made.

***or***

**[2]** Appropriate work is shown, but one conceptual error is made.

***or***

**[2]** Appropriate work is shown to find *TS*, but no further correct work is shown.

**[1]** Appropriate work is shown, but one conceptual error and one computational or rounding error are made.

***or***

**[1]** A correct trigonometric equation is written to find *TS*, but no further correct work is shown.

***or***

**[1]** 1503, but no work is shown.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(34) **[4]** 536, and correct work is shown.

**[3]** Appropriate work is shown, but one computational or rounding error is made.

***or***

**[3]** Appropriate work is shown to find the weight of the base, but no further correct work is shown.

**[2]** Appropriate work is shown, but two or more computational or rounding errors are made.

***or***

**[2]** Appropriate work is shown, but one conceptual error is made.

***or***

**[2]** Correct work is shown to find the volume of the base in cubic feet and/or 85% of the volume in cubic inches, but no further correct work is shown.

**[1]** Correct work is shown to find the volume of the base in cubic inches, but no further correct work is shown.

***or***

**[1]** Appropriate work is shown, but one conceptual error and one computational or rounding error are made.

***or***

**[1]** 536, but no work is shown.

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

## Part IV

For this question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(35) **[6]** A complete and correct proof that includes a concluding statement is written.

**[5]** A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or incorrect.

**[4]** A proof is written that demonstrates a good understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or incorrect.

***or***

**[4]** A proof is written that demonstrates a good understanding of the method of proof, but one conceptual error is made.

**[3]** A proof is written that demonstrates a method of proof, but three statements and/or reasons are missing or incorrect.

***or***

**[3]** $\triangle ABF \cong \triangle CDE$ is proven, but no further correct work is shown.

***or***

**[3]** A proof is written that demonstrates a method of proof, but one conceptual error is made, and one statement and/or reason is missing or incorrect.

**[2]** Some correct relevant statements about the proof are made, but four or more statements and/or reasons are missing or incorrect.

***or***

**[2]** A proof is written with some understanding of the method of proof, but two conceptual errors are made.

**[1]** Only one correct relevant statement and reason are written.

**[0]** The "given" and/or the "prove" statements are rewritten in the style of a formal proof, but no further correct relevant statements are written.

***or***

**[0]** A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

**Map to the Core Learning Standards**
**Geometry**
**June 2018**

| Question | Type | Credits | Cluster |
|---|---|---|---|
| 1 | Multiple Choice | 2 | G-CO.B |
| 2 | Multiple Choice | 2 | G-CO.C |
| 3 | Multiple Choice | 2 | G-CO.A |
| 4 | Multiple Choice | 2 | G-SRT.B |
| 5 | Multiple Choice | 2 | G-SRT.A |
| 6 | Multiple Choice | 2 | G-SRT.C |
| 7 | Multiple Choice | 2 | G-MG.A |
| 8 | Multiple Choice | 2 | G-SRT.C |
| 9 | Multiple Choice | 2 | G-SRT.B |
| 10 | Multiple Choice | 2 | G-GMD.A |
| 11 | Multiple Choice | 2 | G-SRT.B |
| 12 | Multiple Choice | 2 | G-GPE.B |
| 13 | Multiple Choice | 2 | G-CO.C |
| 14 | Multiple Choice | 2 | G-GPE.B |
| 15 | Multiple Choice | 2 | G-GPE.B |
| 16 | Multiple Choice | 2 | G-GMD.B |
| 17 | Multiple Choice | 2 | G-C.A |
| 18 | Multiple Choice | 2 | G-CO.C |
| 19 | Multiple Choice | 2 | G-CO.A |
| 20 | Multiple Choice | 2 | G-GPE.A |
| 21 | Multiple Choice | 2 | G-SRT.B |
| 22 | Multiple Choice | 2 | G-C.B |
| 23 | Multiple Choice | 2 | G-SRT.B |
| 24 | Multiple Choice | 2 | G-SRT.A |
| 25 | Constructed Response | 2 | G-CO.B |
| 26 | Constructed Response | 2 | G-SRT.A |
| 27 | Constructed Response | 2 | G-CO.A |
| 28 | Constructed Response | 2 | G-C.A |
| 29 | Constructed Response | 2 | G-CO.D |
| 30 | Constructed Response | 2 | G-SRT.B |
| 31 | Constructed Response | 2 | G-MG.A |
| 32 | Constructed Response | 4 | G-GPE.B |
| 33 | Constructed Response | 4 | G-SRT.C |
| 34 | Constructed Response | 4 | G-MG.A |
| 35 | Constructed Response | 6 | G-CO.C |